

WEEK 2 STATUS REPORT: ENVIRONMENT SETUP, SYSTEM CONNECTIVITY, & VLANS

1. Virtual Environment & Topology Implementation

- 1 Assembled the physical and logical network blueprint inside the emulation environment workspace using 2911 ISR series routing platforms and 2960 series switching systems.
- 2 Interconnected R1, R2, and R3 physical GigabitEthernet0/0 interfaces straight to the central backplane switch using standard category-5 copper straight-through cabling.

2. Inter-VLAN Routing & Logical Segmentation

- 1 Configured Router-on-a-Stick (ROAS) configurations on R1, R2, and R3 to implement strict network layer isolation.
- 2 Deployed 802.1Q encapsulation parameters via virtualized routing interfaces (**GigabitEthernet0/0.10**) to map cleanly to the management broadcast domain.
- 3 Initialized and built VLAN 10 (Named: 'Management') within the active switch database.
- 4 Assigned switch interface ports FastEthernet0/1 through FastEthernet0/4 explicitly to access mode within VLAN 10, bringing up full connectivity to the automation host.

3. Remote Cryptographic Line Hardening

- 1 Defined specific administrative identity records by modifying default generic naming conventions to unique host identifiers (R1, R2, R3).
- 2 Configured localized system cryptographic parameters by provisioning '**ccp.local**' domain identifiers.
- 3 Generated 1024-bit RSA asymmetrical encryption key pairs on all active routing nodes to authorize SSH processing.
- 4 Enforced secure SSH version 2 (SSHv2), established a 60-second execution idle timeout threshold, and limited handshake retry failures to 3 attempts max.
- 5 Provisioned high-privilege level 15 user access profiles and locked out cleartext inbound protocol access via VTY line terminal isolation (**transport input ssh**).